COL - 761
Data Mining

Team - Rishabh Sama

Rishabh Verma, 2020CS50123
Harshit Kumar, 2020CS10347
Vaibhav Saha, 2020CS10403

20th August, 2023

Algorithm

Algorithm 1: Doing multiple compressions using FP Tree

```
Input: Uncompressed Data
Output: Compressed Data
/* Frequent Itemsets Found USing Standard FP Tree Algorithm */
Freq ← the vector of pairs of Integer(length) and set of strings(items) of Frequent Item set;
NewMap ← the mapping to label new items introduced;
foreach line in Uncompressed or Currently-Compressed File do
    if x of Freq in line then
        x ← −1 ∗ (buffer + index);
        NewMap ← {−1 ∗ (buffer + index), x}
end
return Compressed Data;
;
To Check : x of Freq in line;
Map ← a map of Items in line and Integer(1);
for x in Freq do
    if every item of x in map then
        replace x;
        foreach item in x do
            map.erase(item);
        end
end
```

Now For Multiple Compression:

1. We can run the FP Tree Algorithm on the earlier compressed data with a lower Minimum Support to get somewhat less frequent but frequent itemsets and then we allow these compression (if profitable).

2. For multiple compression, rounds of this step is done until significant compression is achieved.